Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 2 — Quadratics
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第2章 · 二次函数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

P1-C02-R01 · P1 · Chapter 2

The curve C has equation $y = 3x^2 - 12x + 5$.

(a) Express y in the form $a(x - b)^2 + c$. (2)

(b) Hence find

(i) the coordinates of the minimum point of C, (1)

(ii) the exact values of x for which $y = 5$. (2)

(Total for Question 1 is 5 marks)

2. P1-C02-R02 · P1 · Chapter 2

The quadratic equation $x^2 - 2kx + k + 6 = 0$ has equal real roots.

(a) Find the possible values of k. (3)

(b) For each possible value of k, find

(i) the repeated root, (2)

(ii) the corresponding factorised quadratic expression. (2)

(Total for Question 2 is 7 marks)

3.

A rectangle has perimeter 28 cm. Its shorter side has length x cm, where $0 < x \leq 7$.

(a) Show that its area is $49 - (x - 7)^2$ cm^2. (3)

(b) The area is 40 cm^2. Find the dimensions of the rectangle. (3)

(Total for Question 3 is 6 marks)

$f(x) = 2x^2 + 8x + 11$.

(a) By completing the square, show that $f(x) > 0$ for every real x. (2)

(b) Find the exact minimum value of $\frac{1}{2}f(x) - 4$ and the value of x at which it occurs. (3)

(Total for Question 4 is 5 marks)

5.

The graph of $y = x^2 + px + q$ has minimum point $(3, -5)$.

(a) Find p and q. (3)

(b) Find the exact coordinates of the points where the graph meets the x-axis. (2)

(Total for Question 5 is 5 marks)

6. P1-C02-R06 · P1 · Chapter 2

The equation $(m-1)x^2 + 6x + 4 = 0$ has two distinct real roots, and $m \neq 1$.

(a) Show that $m < \frac{13}{4}$. (2)

(b) Find the value of x that solves the equation when $m = 1$, and explain why this case was excluded from part (a). (3)

(Total for Question 6 is 5 marks)

7.

A quadratic expression is $f(x) = x^2 - (k+2)x + 2k$, where k is real.

(a) Factorise $f(x)$ in terms of k. (2)

(b) Given that the graph of $y = f(x)$ has its minimum point on the x-axis, find

(i) the value of k, (2)

(ii) the coordinates of the minimum point. (1)

(Total for Question 7 is 5 marks)

8.

The equation $4x^2 - 4x - 7 = 0$ has roots $\alpha < \beta$.

(a) Find α and β in exact form. (3)

(b) Find the exact distance between the points $(\alpha, 0)$ and $(\beta, 0)$. (2)

(Total for Question 8 is 5 marks)

9.

A rectangular display has area $45\ \text{cm}^2$. Its length is 4 cm greater than its width.

(a) Form a quadratic equation for the width w cm. (2)

(b) Find the dimensions and the perimeter of the display. (3)

(Total for Question 9 is 5 marks)

$y = -2x^2 + 12x - 11$.

(a) Express y in the form $A - B(x - C)^2$. (2)

(b) Hence find

(i) the maximum value of y, (1)

(ii) the exact values of x for which $y = -1$. (2)

(Total for Question 10 is 5 marks)

11. P1-C02-R11 · P1 · Chapter 2

The line $y = c$ meets the curve $y = x^2 - 8x + 13$ at two distinct points.

(a) Find the range of possible values of c. (2)

(b) The two intersections have x-coordinates differing by 6. Find c. (3)

(Total for Question 11 is 5 marks)

$f(x) = x^2 + 6x + k$. The minimum value of $f(x)$ is twice $f(0)$.

(a) Find k. (3)

(b) For this value of k, solve $f(x) = 0$, giving exact answers. (2)

(Total for Question 12 is 5 marks)

13.

A ball is modelled by $h = 1 + 8t - 2t^2$, where h is its height in metres and $t \geq 0$ is time in seconds.

(a) Find the maximum height and the time at which it is reached. (3)

(b) Find the time when the ball first reaches the ground. Give your answer to 3 significant figures. (3)

(Total for Question 13 is 6 marks)

14.

The quadratic expression $x^2 + 2(k-1)x + k^2 + 3$ is defined for real x, k.

(a) Show that its discriminant is $-8(k+1)$. (3)

(b) Find all values of k for which the expression is strictly positive for every real x. (2)

(Total for Question 14 is 5 marks)

The equation $x^2 - 10x + 18 = 0$ has roots p and q, with $p < q$.

(a) Find p and q in exact form. (2)

(b) A new quadratic equation has roots $p + 2$ and $q + 2$. Find this equation in the form $x^2 + ax + b = 0$. (3)

(Total for Question 15 is 5 marks)

A rectangle is inscribed under the line $y = 12 - 2x$ in the first quadrant, with vertices $(0, 0)$, $(x, 0)$, (x, y) and $(0, y)$.

(a) Express its area A in terms of x and state the permitted interval for x. (3)

(b) Without differentiation, find the maximum possible area and the dimensions of the corresponding rectangle. (3)

(Total for Question 16 is 6 marks)

The curve $y = ax^2 + bx + 2$ passes through $(1, 1)$ and $(3, 11)$.

(a) Find a and b. (3)

(b) Find the exact minimum value of y. (2)

(Total for Question 17 is 5 marks)

18.

The equation $kx^2 + 4x + k = 0$ has no real roots. It is given that $k > 0$.

(a) Find the possible values of k. (3)

(b) For $k = 4$, write $kx^2 + 4x + k$ as a completed square and state its minimum value. (3)

(Total for Question 18 is 6 marks)

19.

Given that $x^2 + bx + 12 = (x - r)^2 + s$ for all real x, where $r = \frac{5}{2}$,

(a) find b and s. (3)

(b) Explain why the equation $x^2 + bx + 12 = 0$ has no real solutions. (2)

(Total for Question 19 is 5 marks)

A quadratic equation is $x^2 + px + q = 0$. One root is 2 and the other root is 3 greater.

(a) Find p and q. (3)

(b) For these values of p, q, find the value of c such that $x^2 + px + q = c$ has equal real roots. (2)

(Total for Question 20 is 5 marks)

21. P1-C02-R21 · P1 · Chapter 2

The graph of $y = x^2 - 4x + d$ meets the positive y-axis and has two distinct positive x-intercepts.

(a) Find the range of possible values of d. (3)

(b) Given that the smaller x-intercept is 1, find the other intercept and d. (3)

(Total for Question 21 is 6 marks)

A quadratic function has minimum value -8 at $x = -1$ and passes through $(1, 4)$.

(a) Find its equation in the form $y = ax^2 + bx + c$. (3)

(b) Find the exact values of x for which $y = 0$. (3)

(Total for Question 22 is 6 marks)

23. P1-C02-R23 · P1 · Chapter 2

For $x \geq 0$, $C = 2x^2 - 20x + 61$ is a model for the daily cost, in pounds, of producing a batch of size x hundred items.

(a) Find the minimum value of C and the batch size at which it occurs. (3)

(b) Find the two batch sizes for which $C = 29$. (3)

(Total for Question 23 is 6 marks)

The expression $f(x) = x^2 + 2ax + 9$ has minimum value 5.

(a) Find the possible values of a. (3)

(b) For each possible value of a, find

(i) the coordinates of the minimum point, (1)

(ii) the number of real solutions of $f(x) = 4$. Give a reason. (2)

(Total for Question 24 is 6 marks)

$x^2 - 6x + 7 = 2(x - 1)$.

(a) Solve the equation, giving exact answers. (3)

(b) Find the distance between the midpoint of the two roots on the number line and the origin. (2)

(Total for Question 25 is 5 marks)

26.

The quadratic equation $x^2 + (k - 4)x + 4 = 0$ has equal real roots.

(a) Find the two possible values of k. (3)

(b) For which of these values is the repeated root negative? State the repeated root. (2)

(Total for Question 26 is 5 marks)

A garden has width x m and length $(x + 6)$ m. A path of width 1 m runs inside all four edges. It is given that $x > 2$.

(a) Show that the remaining planted area is $(x^2 + 2x - 8)\ \text{m}^2$. (3)

(b) The planted area is $40\ \text{m}^2$. Find the exact outer dimensions of the garden. (3)

(Total for Question 27 is 6 marks)

The equation $3x^2 + 6x + c = 0$ has roots $-1 - r$ and $-1 + r$, where $r > 0$.

(a) Express c in terms of r. (2)

(b) Given that $c = -24$, find

(i) the value of r, (2)

(ii) the two roots of the equation. (1)

(Total for Question 28 is 5 marks)

29. P1-C02-R29 · P1 · Chapter 2

A curve has equation $y = 2x^2 - 4kx + k^2 + 6$.

(a) Write y in completed-square form. (2)

(b) The curve touches the x-axis. Find the two possible values of k and the coordinates of the point of contact in each case. (3)

(Total for Question 29 is 5 marks)

30.

The height of an arch is modelled by $h = 12 - \frac{1}{3}(x - 6)^2$ metres, for $0 \leq x \leq 12$, where x is horizontal distance from its left end.

(a) Find the width of the arch at ground level and its maximum height. (3)

(b) A horizontal beam is fitted at height 9 m. Find

(i) the exact horizontal coordinates of the two ends of the beam, (2)

(ii) the length of the beam. (1)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C02-R01 · 5 marks

(a) · 2 marks

$y = 3(x-2)^2 - 7$.

提出3后配方$x^2 - 4x = (x-2)^2 - 4$。

M1	正确配方
A1	三个常数

(b) Hence find

(i) · 1 marks

$(2, -7)$.

平方项最小为0，在x=2时取得。

B1	正确最小点

(ii) · 2 marks

$x = 0, 4$.

$3(x-2)^2 = 12$，取两个平方根得x−2=±2。

M1	建立平方关系
A1	两根

Question 2

P1-C02-R02 · 7 marks

(a) · 3 marks

$k = -2, 3$.

判别式为$4k^2 - 4(k+6) = 4(k-3)(k+2)$；令其为$0$。

M1	判别式等于零
M1	正确解参数方程
A1	两个k值

(b) For each possible value of k, find

(i) · 2 marks

$x = -2$ when $k = -2$; $x = 3$ when $k = 3$.

重根为$-b/(2a) = k$。

M1	求重根
A1	与两个k正确配对

(ii) · 2 marks

$(x+2)^2$ and $(x-3)^2$, respectively.

分别把重根写成对应平方因式；展开可复核常数项。

M1	根据重根写平方
A1	两式

Question 3

P1-C02-R03 · 6 marks

(a) · 3 marks

$A = x(14 - x) = 49 - (x - 7)^2$.

另一边为14−x；展开再配方。

M1	正确另一边
M1	面积关系
A1	配方结果

(b) · 3 marks

4 cm by 10 cm.

$(x - 7)^2 = 9$给x=4或10；短边限制只允许4。

M1	解平方关系
A1	按范围取4
A1	两边及单位

Question 4

P1-C02-R04 · 5 marks

(a) · 2 marks

$f(x) = 2(x+2)^2 + 3 \geq 3 > 0.$

平方非负，且常数下界为严格正数3。

M1	正确配方
A1	利用下界证明严格正

(b) · 3 marks

Minimum $-\frac{5}{2}$, at $x = -2$.

乘以正数1/2不改变最小值位置；最小值$3/2 - 4$。

M1	使用f的最小值
A1	新最小值
B1	取得位置

Question 5

P1-C02-R05 · 5 marks

(a) · 3 marks

$p = -6,\ q = 4.$

写成$y = (x-3)^2 - 5 = x^2 - 6x + 4$，比较系数。

M1	使用顶点式
A1	p
A1	q

(b) · 2 marks

$(3 - \sqrt{5}, 0)$ and $(3 + \sqrt{5}, 0).$

令y=0，得$(x-3)^2 = 5$；两根都应写成坐标。

M1	解配方等于零
A1	两个坐标

Question 6

P1-C02-R06 · 5 marks

(a) · 2 marks

$36-16(m-1)>0 \Rightarrow m<13/4$.

二次方程两不同实根要求判别式严格大于0。

M1	严格判别式条件
A1	参数上界

(b) · 3 marks

$x=-\frac{2}{3}$; the equation is linear, not quadratic.

m=1时二次项消失，只剩6x+4=0，所以不适用二次方程两根的讨论。

M1	解退化后方程
A1	正确根
B1	解释二次项消失

Question 7

P1-C02-R07 · 5 marks

(a) · 2 marks

$f(x) = (x-2)(x-k)$.

展开右式可得一次项$-(k+2)x$和常数$2k$。

M1	识别两个因式
A1	正确完整分解

(b) Given that the graph of $y = f(x)$ has its minimum point on the x-axis, find

(i) · 2 marks

$k = 2$.

最小点在x轴意味着两根相等，$2=k$。

M1	联系最小点与重根
A1	k为2

(ii) · 1 marks

$(2, 0)$.

此时$f(x) = (x-2)^2$。

B1	最小点坐标

Question 8

P1-C02-R08 · 5 marks

(a) · 3 marks

$\alpha = \frac{1}{2} - \sqrt{2}, \beta = \frac{1}{2} + \sqrt{2}.$

用二次公式，判别式16+112=128，$\sqrt{128} = 8\sqrt{2}$。

M1	使用二次公式
M1	化简根式
A1	排序正确的两根

(b) · 2 marks

$2\sqrt{2}.$

两点同在x轴，距离为β−α。

M1	使用有序根之差
A1	正确精确距离

Question 9

P1-C02-R09 · 5 marks

(a) · 2 marks

$w^2 + 4w - 45 = 0$.

长度w+4，面积关系$w(w+4)=45$。

M1	建立面积式
A1	标准二次方程

(b) · 3 marks

5 cm by 9 cm; perimeter 28 cm.

方程分解为(w+9)(w−5)=0；长度必须正，取w=5。

M1	正确解两代数根
A1	排除负宽并求长
A1	周长及单位

Question 10

P1-C02-R10 · 5 marks

(a) · 2 marks

$y = 7 - 2(x-3)^2$.

提出−2并配方。

M1	正确处理负系数配方
A1	结果

(b) Hence find

(i) · 1 marks

7.

被减去的平方项非负，在x=3为零。

B1	最大值7

(ii) · 2 marks

$x = 1, 5$.

$7 - 2(x-3)^2 = -1$，故平方为4。

M1	正确平方关系
A1	两解

Question 11

P1-C02-R11 · 5 marks

(a) · 2 marks

$c > -3$.

曲线为$(x-4)^2-3$，水平线高于最小值才有两个交点。

M1	求最小值或判别式
A1	严格范围

(b) · 3 marks

$c = 6$.

两横坐标为$4 \pm \sqrt{c+3}$，差为$2\sqrt{c+3} = 6$。

M1	建立根间距关系
M1	解平方关系
A1	c为6

Question 12

P1-C02-R12 · 5 marks

(a) · 3 marks

$k = -9$.

配方得最小值k−9；f(0)=k，因此k−9=2k。

M1	正确最小值
M1	建立条件方程
A1	k

(b) · 2 marks

$x = -3 \pm 3\sqrt{2}$.

此时$f = (x+3)^2 - 18$，解平方方程。

M1	正确代入k并配方
A1	两个根

Question 13

P1-C02-R13 · 6 marks

(a) · 3 marks

9 m at $t = 2$ s.

$h = 9 - 2(t-2)^2$，最大时平方项为零。

M1	配方
A1	最大高度
B1	对应时间

(b) · 3 marks

$t = 2 + \frac{3}{2}\sqrt{2} \approx 4.12$ s.

令h=0得$2t^2 - 8t - 1 = 0$，两根$2 \pm 3\sqrt{2}/2$；只保留非负时间。

M1	设高度为0并解二次方程
A1	选物理根
A1	按要求近似及单位

Question 14

P1-C02-R14 · 5 marks

(a) · 3 marks

$\Delta = 4(k-1)^2 - 4(k^2+3) = -8k - 8.$

展开平方后k^2项抵消。

M1	正确判别式
M1	展开化简
A1	目标式

(b) · 2 marks

$k > -1.$

二次项系数为正，恒严格正要求判别式负；$k=-1$时有零点不满足严格正。

M1	使用开口向上与判别式负
A1	严格范围

Question 15

P1-C02-R15 · 5 marks

(a) · 2 marks

$p = 5 - \sqrt{7}$, $q = 5 + \sqrt{7}$.

配方为$(x-5)^2=7$。

M1	配方
A1	有序两根

(b) · 3 marks

$x^2 - 14x + 42 = 0$.

新根为$7 \pm \sqrt{7}$，故$(x-7)^2 - 7 = 0$，展开。

M1	写出新根
M1	建立配方方程
A1	正确标准式

Question 16

P1-C02-R16 · 6 marks

(a) · 3 marks

$A = 12x - 2x^2$, $0 < x < 6$.

面积xy；两边长必须严格正，要求x>0与12−2x>0。

M1	正确面积式
A1	面积多项式
B1	开区间

(b) · 3 marks

18 square units; width 3, height 6.

$A = 18 - 2(x-3)^2$，顶点在x=3且位于允许范围。

M1	面积配方
A1	最大面积
A1	两个尺寸

Question 17

P1-C02-R17 · 5 marks

(a) · 3 marks

$a=2, b=-3.$

代入得a+b=−1及9a+3b=9；消元得a=2再求b。

M1	两个条件方程
M1	消元
A1	两系数

(b) · 2 marks

$\frac{7}{8}$.

$2x^2-3x+2=2(x-3/4)^2+7/8$。

M1	正确配方
A1	最小值7/8

Question 18

P1-C02-R18 · 6 marks

(a) · 3 marks

$k > 2$.

$16 - 4k^2 < 0$，结合k正得k>2。

M1	判别式小于零
M1	使用正值条件
A1	范围

(b) · 3 marks

$4(x + \frac{1}{2})^2 + 3$; minimum 3.

代入k后提出4配方，常数项4−1=3。

M1	正确配方
A1	完整式
B1	最小值

Question 19

P1-C02-R19 · 5 marks

(a) · 3 marks

$b = -5$, $s = \frac{23}{4}$.

比较一次项b=−2r；常数比较12=r²+s。

M1	比较系数
A1	b
A1	s

(b) · 2 marks

Its minimum value is $23/4 > 0$.

由(a)配方式，平方非负，整个式子至少23/4，不可能等于0。

M1	使用配方的非负平方
A1	正确的无根理由

Question 20

P1-C02-R20 · 5 marks

(a) · 3 marks

$p = -7,\ q = 10.$

另一根5；对应$(x-2)(x-5) = x^2 - 7x + 10$。

M1	得出第二根并写乘积
A1	p
A1	q

(b) · 2 marks

$c = -\frac{9}{4}.$

配方为$(x - 7/2)^2 - 9/4$，水平线仅在最小值处给重根。

M1	配方或判别式等零
A1	c

Question 21

P1-C02-R21 · 6 marks

(a) · 3 marks

$0 < d < 4$.

y轴正截距给d>0；判别式16−4d>0给d<4；两根$2 \pm \sqrt{4-d}$均正。

M1	正截距条件
M1	不同实根条件
A1	完整范围并保证正根

(b) · 3 marks

Other intercept $(3, 0)$; $d = 3$.

代入x=1得1−4+d=0；方程为(x−1)(x−3)=0。

M1	代入已知根
A1	d
A1	第二截距

Question 22

P1-C02-R22 · 6 marks

(a) · 3 marks

$y = 3x^2 + 6x - 5.$

顶点式$y = a(x+1)^2 - 8$；代入(1,4)得4a−8=4所以a=3。

M1	正确顶点式
M1	用点求a
A1	展开标准式

(b) · 3 marks

$x = -1 \pm \dfrac{2\sqrt{6}}{3}.$

由顶点式得$(x+1)^2=8/3$，开方并化简根式。

M1	正确平方关系
M1	化简根式
A1	两根

Question 23

P1-C02-R23 · 6 marks

(a) · 3 marks

£11, for 500 items.

$C = 2(x-5)^2 + 11$；x=5表示500件。

M1	配方
A1	最小费用
A1	正确解释百件单位

(b) · 3 marks

200 items and 800 items.

$2(x-5)^2 = 18$得x=2或8，再换算件数。

M1	解平方方程
A1	两个x值
A1	两个实际批量

Question 24

P1-C02-R24 · 6 marks

(a) · 3 marks

$a = \pm 2$.

$f = (x+a)^2 + 9 - a^2$；$9-a^2=5$。

M1	正确最小值
M1	参数平方关系
A1	两个值

(b) For each possible value of a, find

(i) · 1 marks

$(-2, 5)$ if $a = 2$; $(2, 5)$ if $a = -2$.

最小点横坐标为$-a$，纵坐标均5。

B1	两个正确配对的顶点

(ii) · 2 marks

No real solutions in either case.

f的最小值5大于4，水平线$y=4$不与曲线相交。

M1	比较最小值与4
A1	无实根结论

Question 25

P1-C02-R25 · 5 marks

(a) · 3 marks

$x = 4 \pm \sqrt{7}$.

整理成$x^2-8x+9=0$，配方$(x-4)^2=7$。

M1	正确移项
M1	配方或公式
A1	两根

(b) · 2 marks

4.

两根关于4对称，其中点为4，离原点距离4。

M1	求两根中点
A1	正确距离

Question 26

P1-C02-R26 · 5 marks

(a) · 3 marks

$k = 0, 8.$

判别式$(k-4)^2-16=0$，故$k-4=\pm 4$。

M1	判别式等零
M1	解平方方程
A1	两k值

(b) · 2 marks

$k = 8$, repeated root $x = -2.$

重根为$(4-k)/2$；$k=0$给2，$k=8$给-2。

M1	分别求重根
A1	选择k及根

Question 27

P1-C02-R27 · 6 marks

(a) · 3 marks

$A=(x-2)(x+4)=x^2+2x-8.$

内部两尺寸各减2，分别为x−2和x+4。

M1	正确内侧尺寸
M1	建立面积乘积
A1	展开结果

(b) · 3 marks

Width 6 m, length 12 m.

$x^2+2x-48=0$给$(x+8)(x-6)=0$；按$x>2$取6。

M1	正确二次方程
M1	因式分解并选根
A1	外尺寸

Question 28

P1-C02-R28 · 5 marks

(a) · 2 marks

$c = 3 - 3r^2$.

将方程写为$3(x+1)^2+c-3=0$；在根处平方为r^2。

M1	正确配方或因式乘积
A1	c表达式

(b) Given that $c = -24$, find

(i) · 2 marks

$r = 3$.

$-24=3-3r^2$给$r^2=9$，用r正取3。

M1	解平方关系
A1	r值

(ii) · 1 marks

$x = -4, 2$.

代入$-1\pm r$。

B1	两根

Question 29

P1-C02-R29 · 5 marks

(a) · 2 marks

$y = 2(x-k)^2 + 6 - k^2$.

前两项配为$2[(x-k)^2-k^2]$，与原k^2合并。

M1	正确含参数配方
A1	常数项

(b) · 3 marks

$k=\sqrt{6}$: $(\sqrt{6},0)$; $k=-\sqrt{6}$: $(-\sqrt{6},0)$.

接触x轴意味着最小值$6-k^2=0$，顶点横坐标k。

M1	令最小值为0
A1	两参数
A1	两接触点

Question 30

P1-C02-R30 · 6 marks

(a) · 3 marks

Ground width 12 m; maximum height 12 m.

h=0给$(x-6)^2=36$，端点x=0和12；配方形式显示最高12。

M1	求两个地面位置
A1	宽度
B1	最大高度

(b) A horizontal beam is fitted at height 9 m. Find

(i) · 2 marks

$x = 3, 9$.

$12-(x-6)^2/3=9$，故平方为9。

M1	建立平方方程
A1	两坐标

(ii) · 1 marks

6 m.

右端位置减左端位置9−3。

B1	长度6米